Representing Visible Difference:

Extraordinary Paleness of Hair, Skin and Unusual Eyes in English Writing c700-1650 with comparisons up to 1914

Research outline - Lee Goddard

1. Research problem and rationale

This project asks how extraordinary paleness of hair, skin and eyes has been described and interpreted in English writing, and how meanings ranging from the exalted and sacred to the monstrous and pathological became attached to it in different historicfal contexts. I use "extraordinary paleness" as a deliberately broad, pre-medical descriptive term rather than as synonym for the later medical category of albinism that captures some but not all of the people and descriptions concerned. The project traces descriptions and interpretations in English writing from the Anglo-Saxon period to the early twentieth century, and treats the apparent shift as a question to be tested rather than a story to be assumed.

Thomas White's 2011 thesis, *Their Whiteness Is Not Like Ours: A Social and Cultural History of Albinism and Albino Identities, 1650-1914*, documents the construction of albinism as a scientific, medical and social category, and the ways in which people with albinism have managed that identity. It does not examine earlier English descriptions, and does not investigate whether the negative associations attached to albinism originated with its medical classification, or whether medical discourse formalised and perpetuated much older cultural associations.

These physical characteristics have historically been open to a multiplicity of interpretations that include the natural-philosophical, theological, symbolic and literary. The project will investigate what meanings and cultural values were attached to this visible difference in earlier periods, how those meanings changed, and whether some interpretative possibilities were diminished or lost as the phenomenon came to be understood in medical terms.

The principal investigation concerns English material from the Anglo-Saxon period to around White's starting point of 1650. It asks whether this material shows a plural repertoire of meanings, drawing on biblical, exegetical and other traditions, that cannot be adequately explained through the later history of albinism, and whether that repertoire later narrowed. The later period is comparative: one chapter will set these findings against White's account and, time permitting, consider Gothic fiction and early science fiction that White does not address, as a way of asking how far medicalised understandings contributed to pejorative associations.

2. Research questions and hypotheses

The central question is: **How were extraordinary degrees of bodily paleness, possibly with unusual eyes, described, interpreted, transmited and perhaps transformed and displaced, in English writing between approximately 700 and 1914, and how did the meanings attached to these characteristics change over that period?**

My initial hypothesis is that English writing before 1650 contains a plural repertoire of meanings for extraordinary paleness of hair, skin and unusual eyes, some exalted or sacred, some ambiguous, some monstrous, and that biblical and exegetical traditions were an important source of the exalted strand. The

dominance of pathological and villainous readings in later writing would then be a narrowing of this repertoire rather than a continuation of it. The hypothesis would be weakened if the earlier material proved consistently negative, or if exalted readings attached only to divine figures as symbol and never to physical bodies.

A second question is whether later naturalistic and medical descriptions drew on this earlier material, or developed largely through other traditions, including classical authorities, travel writing and medical literature. Neither continuity nor discontinuity will be assumed: both are questions for investigation.

3. Biblical and exegetical foundations

The descriptions in Daniel 7:9 and Revelation 1:13-14 provide an important starting point. Daniel describes the Ancient of Days with hair like pure wool, while Revelation describes the hair of the Son of Man as white like wool or snow, alongside striking features of the eyes. The language of whiteness, brightness and radiance is not necessarily interchangeable. The relevant Aramaic, Greek and Latin terms, together with their place in the Hebrew literary tradition and their subsequent Anglo-Saxon, Anglo-Norman and English transmission, need to be established in their original linguistic and interpretative contexts and considered separately.

These figures are divine or exalted, and the project does not assume that their whiteness was ever read simply as bodily difference. Whether exegetes and vernacular writers treated the description as a physical marker, an allegory of eternity or wisdom, or something else is itself a question for investigation. Typological exegesis also links the Ancient of Days with the Son of Man, a relationship that shapes how the imagery was received. Comparative Jewish and early Christian material, such as the description of Noah's birth in 1 Enoch 106, may show the imagery read as bodily difference; how far such material reached English writing is a question of transmission.

The project will examine the distinctions between terms denoting whiteness, brightness or radiance, and snow, and how these were preserved, altered or obscured through translation and commentary. The exegetical traditions of Jerome, Origen and Cyril of Jerusalem show that these descriptions could sustain different interpretations. Bede's treatment is especially relevant to their reception in the English medieval context; notably, his interpretation of the Son of Man's appearance does not simply equate white hair with old age. Examining these traditions alongside the source languages will help establish how far white hair could carry meanings beyond that of a marker of age.

These materials will be examined not simply as sources of literary imagery, but as evidence for historically available ways of interpreting extraordinary physical appearance.

4. Corpus and historical scope

The principal corpus comprises English writing from the Old English period to 1650. Later material is comparative to White, as set out at the end of this section.

Old English. The surviving evidence is sufficiently limited to make a systematic examination feasible. Using the *Dictionary of Old English Corpus* and the *Thesaurus of Old English* (for semantic fields), along with C. P. Biggam's work on the period's colour terms, I aim to identify and examine all occurrences of white hair, together with potentially relevant descriptions of extraordinary bodily paleness and unusual eyes. This will involve searching (manually and computationally) for terms denoting whiteness, brightness and related physical characteristics, followed by examination of each occurrence in context. *The Wonders of the East*, descriptions of Grendel's eyes, and terms such as *ælfscȳne* provide initial points of investigation.

Comparative material. Alaric Hall's work on elves in Anglo-Saxon England connects *ælfscȳne* with wider Germanic beliefs. Old Norse descriptions of whiteness may provide useful parallels, although such material will be considered only where it helps illuminate the English evidence. Old Welsh and Old Irish material is sadly likely to lie beyond the scope of the project.

Middle English. Period biblical and exegetical material, the investigation will use the *Middle English Dictionary* and the *Corpus of Middle English Prose and Verse* to examine descriptions of extraordinary physical appearance in vernacular literature and in encyclopaedic and natural-historical texts, including Bartholomaeus Anglicus's *De proprietatibus rerum*, especially the section *De Albania*, and John Trevisa's translation.

Early modern. EEBO (or EEBO-TCP) will be the main source. Orthographic variation and OCR noise will be handled computationally, as described in Section 6.

Later comparison (1650-1914). This material builds on White's research and examines selected developments in naturalistic, descriptive and medical accounts, together with selected Gothic and early science fiction (for example H. G. Wells' *Invisible Man*), in which extraordinary pallor is sometimes associated with monstrosity, degeneration and otherness. It will help explore the relationship between medicalised understandings and the persistence or transformation of older cultural associations. The purpose is not to reproduce White's history, but to establish how the earlier material relates to the emergence and subsequent use of the category of albinism. Should time require, the Gothic and science fiction strand can be reduced to a short comparative section or epilogue without affecting the central argument.

5. Method and interpretation

Close textual and philological analysis will focus on works identified through existing scholarship and preliminary research, supplemented by passages and texts discovered through computational retrieval across larger historical corpora. This combination of historically informed selection and semantic search will help identify relevant material beyond the limits of familiar terminology or previously recognised examples.

Individual passages will be interpreted in their literary, theological, historical and generic contexts, drawing on both New Historicist and cultural-materialist approaches, particularly Dollimore's work on the cultural and ideological construction of deviance, and Ricoeur's writing on metaphor, symbolism and interpretation. The transmission and reinterpretation of biblical material will be considered in the light of Fabiny and Frye's work on typology and biblical reception. Particular attention will be paid to quotation, allusion and the meanings attached to unusual physical characteristics, as well as to the responses they elicit from other characters or observers.

The investigation will distinguish extraordinary bodily appearance from more conventional uses of white as a symbolic or aesthetic colour. Ordinary fair complexions, age-related hair colours, and descriptions of white clothing or materials will provide controls. Other unusual bodily characteristics, including descriptions of giants and dwarfs, will help establish whether particular features are being treated as exceptional.

The surviving evidence is uneven, and the project will not assume that every description can be assigned a single meaning. An important part of the analysis will be to identify ambiguity, competing interpretations and changes in emphasis, rather than to impose a continuous history on superficially similar descriptions.

6. Computational approach

I have been developing a computational search system to help identify relevant passages across large historical collections. It currently combines historical text ingestion, passage-level semantic retrieval, chronological filtering and source-level provenance. Its embeddings are based on MacBERTh (Manjavacas and Fonteyn 2021), a historically pretrained language model developed for English texts from 1450–1950, including material from EEBO and ECCO. Its suitability for the project, particularly for earlier material, has been observed in preliminary work but will nonetheless be tested rather than assumed.

The search strategy works in two directions: forward from known early descriptions and biblical passages, identifying semantically related material in later writing; and backwards from later terminology associated with albinism, including *albino*, to identify earlier passages that might otherwise be missed because they use different vocabulary. This should help locate both potential continuities and apparently unrelated descriptions.

Computational retrieval will be used for discovery rather than interpretation, allowing a much larger body of material to be surveyed than would otherwise be feasible. Lexical search will provide a baseline against which the effectiveness of semantic retrieval can be assessed, while known passages will provide an initial test set. The project will also investigate the suitability of MacBERTh and other historical language models for the earlier material, and the potential benefits of adapting or further training such models on Old and Middle English texts. Retrieved passages from all periods will be subjected to close textual and historical analysis, with particular attention to the linguistic and interpretative challenges of earlier forms of English.

7. Contribution and significance

The project will establish a longer history of how extraordinary paleness was represented and understood in English writing, particularly before the emergence of albinism as a medical category. It will investigate whether earlier theological, literary and cultural interpretations offered meanings that were subsequently transformed, marginalised or lost, and how far later accounts represent a break with, or continuation of, those traditions. In doing so it will contribute to medieval literary history, biblical reception and the history of the body.

The project also has a contemporary significance. Although albinism is widely estimated to affect around one person in 17,000 in the UK, people with albinism remain relatively invisible in mainstream culture, and when they are represented it is often in association with villainy, monstrosity or otherness. By recovering a more complex history of the ways in which extraordinary paleness has been depicted and interpreted, the project hopes to challenge the assumption that such associations are natural or inevitable, and to provide a historical basis for understanding and questioning their persistence.

The project will demonstrate the potential of computational semantic retrieval as a tool for historical literary research, particularly for identifying related material across periods of linguistic and conceptual change. It takes up the question posed by McGillivray et al. (2024), *Why Does Digital History Need Diachronic Semantic Search?*, by exploring how such methods can support the investigation of changing cultural concepts and representations across historical corpora.

8. Preparation and supervision

I studied English at Sussex (BA Hons 2:1, 1999) before completing an EPSRC-funded one-year MSc in Knowledge-based Systems at COGS in 2000, and have since worked for more than twenty-five years as a consultant software developer, including on large-scale search, data integration and information retrieval. I have developed a working prototype for the computational component (Section 6).

The digital humanities aspects of the project have been discussed with Barbara McGillivray at King's College London, who has indicated her willingness to provide co-supervision in that area. I am therefore particularly interested in exploring whether the project could be supported by a literary supervisor with expertise in medieval English literature, biblical reception and the interpretation of earlier texts.

I have lived with albinism for over fifty years, a position that informs my interest and my awareness of assumptions in modern descriptions of visible difference. I will treat it as a stated position rather than a premise: the sources, not the motivation, will determine the conclusions.

Select bibliography

- Bede (2001). *Expositio Apocalypseos*. Edited by Roger Gryson. Corpus Christianorum Series Latina 121A. Turnhout: Brepols.
- Bede (2013). *Commentary on Revelation*. Translated by Faith Wallis. Translated Texts for Historians 58. Liverpool: Liverpool University Press.
- Biggam, C. P. (2012) *The Semantics of Colour: A Historical Approach*. Cambridge: Cambridge University Press.
- Crawford, Jackson (2014). *The Historical Development of Basic Color Terms in Old Norse-Icelandic.* PhD thesis, University of California, Los Angeles.
- Dollimore, Jonathan (1984). *Radical Tragedy: Religion, Ideology and Power in the Drama of Shakespeare and His Contemporaries*. Brighton: Harvester Press.
- Fabiny, Tibor (2009). *Typology: Pros and Cons in Biblical Hermeneutics and Literary Criticism (from Leonhard Goppelt to Northrop Frye).* Rilce-Revista De Filologia Hispanica. [https://doi.org/10.15581/008.25.26308].
- Frye, Northrop (1982). *The Great Code: The Bible and Literature*. New York: Harcourt Brace Jovanovich.
- Hall, Alaric (2007). *Elves in Anglo-Saxon England: Matters of Belief, Health, Gender and Identity*. Anglo-Saxon Studies 8. Woodbridge: Boydell Press.
- Manjavacas, Enrique, and Lauren Fonteyn (2021). *MacBERTh: Development and Evaluation of a Historically Pre-trained Language Model for English (1450-1950)*. In *Proceedings of the Workshop on Natural Language Processing for Digital Humanities*, pages 23-36, NIT Silchar, India. NLP Association of India (NLPAI).
- McGillivray, Barbara, Federico Nanni, and Kaspar Beelen (2024). *Why Does Digital History Need Diachronic Semantic Search?* in *Computational Humanities*.
- Poleg, Eyal (2013). *Approaching the Bible in Medieval England*. Manchester: Manchester University Press.
- Ricoeur, Paul (1969). *The Symbolism of Evil*. Boston: Beacon Press.
- White, Thomas (2011). *"Their Whiteness Is Not Like Ours": A Social and Cultural History of Albinism and Albino Identities, 1650-1914.* PhD thesis, University of Manchester.